2

Theoretical Underpinnings: Property

This chapter sets out a basic conceptual foundation for this book, which will be used in the case study chapters to determine the fundamental question of the nature of assets exemplified in this book. We will consider whether certain types of digital assets fit within the notion of property and consequently transmit on death like some of the more traditional types of property.

Suppose digital assets analysed in this book are found to be property or protected by IP rights (copyright in particular). In that case, their transmission is clear: property transmits on death, and copyright protection lasts for many years post-mortem (seventy years in the referent jurisdictions, US and UK). To confirm or discard this proposition for each of the case studies, we will explore the meaning of property, its incidents and justifications in this chapter. This discussion will be further applied to examine whether social networks, emails, games and virtual world asserts are property objects.

The analysis sits alongside the discussion in Chapter 3, where autonomy and privacy are examined as further examples of the conceptual underpinning of the book. I, however, do not rely on any claim about the relationship between property, privacy and autonomy. I make no claim, for example, that autonomy underpins property or that the concept of privacy somehow 'mediates' between the concepts of autonomy and property. Instead, property is examined from various angles simply to determine whether virtual assets could (doctrinal question) and should (normative question) be regarded as property. If the answer to either of these questions is 'yes', then these assets can be transmitted on death by testate and intestate succession laws. If the answer is 'no', because some assets are primarily made of information and personal data that cannot constitute property, privacy and autonomy take precedence in this conceptual framework.

This chapter will not discuss copyright from a theoretical perspective. The reason for this is that copyright's transmission on death is clear and settled in law, so if a digital asset meets copyright protection requirements, this right will be passed on to the heirs of a deceased user. Instead, the chapters discussing case studies will assess whether some of their content satisfies copyright requirements from a doctrinal perspective. Although the transmission of property is also evident, property features are not as settled or harmonised in law or theory. Conversely, the requirements for copyright protection have been harmonised (at least from a black letter law perspective, without going into the debate around whether the current copyright regime is desirable or justified). Instead, as will be seen later in this chapter, it is highly contested whether the information or other non-copyrightable content of digital assets is or should be property. Therefore, we will debate property from a theoretical and doctrinal perspective to enable this evaluation in the case study chapters.

Conversely, as another dominant type of digital asset content, information has not been assessed from the perspective of post-mortem transmission. The legal nature of different types of information is often unclear, and one of the most significant considerations is whether information can be considered property. If the answer is affirmative, either from a black letter law or a normative perspective, then the content including predominantly information will transmit or should transmit on death like traditional property. Conversely, if the answer is negative or unclear, each case study in this book will be assessed from the perspective of its specific content, excluding the general transmission of information. The discussion on property in information and personal data is more specific than the general theorising of property set out in this chapter, and it is primarily relevant to the case study analyses in Chapters 4 and 6. Therefore, that specific discussion will utilise findings of the general concept of property explored in this chapter but will be delayed to Chapter 6, where it sits more naturally.

2.1 Defining the Notion of Property

This section will briefly explore the origin, usage and possible definitions of the word 'property'. It will demonstrate that all attempts at a comprehensive and all-encompassing definition, even on a fundamental, abstract level, are fruitless. This discussion is necessary for this book since digital assets are typically located in a transnational space, for example on a server physically located in Ireland, owned by a US company, and accessed by and created by users from many different jurisdictions. Therefore, issues relating to the transmission of digital assets are likely to be even more complex when systems have wildly divergent views on what constitutes property.

George notes that property is 'notoriously difficult to define', and the debate about its definition and nature has been going on for ages and 'seems set to rage for some time yet'.[1] Throughout this chapter, the key features of the property concept will be explored and suggested, even if it does not lead to a precise conclusion about the definition and nature of this critical concept.

The word 'property' comes from the Latin *proprietas*, possibly through French *propriété*, meaning 'the peculiar nature or quality of a thing' and 'ownership'. The word is derived from the adjective *proprius* meaning 'own' or 'peculiar', opposed to *communis* (common) or *alienus* (another's). Furthermore, the word can be rooted in the Greek *προ* or Sanskrit *pra* meaning 'in front of', 'before', 'close to', 'on behalf of'. Donahue interprets this core meaning as something that represented an idea of 'what distinguishes an individual or a thing from a group or another' even before getting its legal meaning.[2]

The word has been used in different contexts and with various and sometimes contradictory meanings between the different legal systems and within the same ones. For instance, older usage in England referred to a relationship between a persona and resources, while later, after the seventeenth century, the meaning pertained to the object of an ownership interest.[3] Along this line, Gray indicates a blurred distinction between property as a 'relationship' and property as a 'thing', arguing that the former use is correct and quoting Bentham and Macpherson to support his stance.[4] Further, he emphasises the dynamic quality of this relationship, changing in time in both subjects and objects of property.

It would be highly complex and, arguably, impossible to provide a widely accepted definition of property for all legal cultures and systems. This is a fact even in a philosophical, abstract definition, since throughout the history of contemplation on property there has been hardly any agreement on its essence and definition. There have been attempts to offer some common characteristics for the notion. For example, as Honoré puts it, ownership, *dominium*, *propriété* and *Eigentum* 'stand not merely for the greatest interest in a thing in a particular system but for a type of interest with common features transcending particular systems'.[5] Honoré uses the term in 'the "liberal" concept of "full" individual ownership'.[6]

1 A George, 'The difficulty of defining "property"' (2005) 25(4) *Oxford Journal of Legal Studies* 813, 813.

2 See J Donahue, 'The future of property predicted from its past' in JR Pennock and JW Chapman (eds), *Property* (New York University Press 1980) 31.

3 JW Harris, *Property and Justice* (Clarendon Press 1996) 10.

4 K Gray, *Elements of Land Law* (Butterworths 1987) 8–14.

5 AM Honoré, 'Ownership' in AG Guest (ed.), *Oxford Essays in Jurisprudence: A Collaborative Work* (Oxford University Press 1961) 108.

6 Ibid. 107.

Definitions in legal codes, according to Honoré, are not 'a safe guide'. However, even though he notes the similarity between the French Civil Code and Soviet Civil Code definitions, both emphasising the absoluteness in the term, he also warns of the different limits laid down by law.[7] According to this, it could be argued that ownership is the greatest interest in a thing in many modern legal systems, but the argument about the common features is much harder to sustain.

Another notable example in support of the argument put forward by Honoré is Blackstone's view on property, which is similar to that predominantly accepted in the Continental, civil law tradition. Blackstone uses the term 'the right of property', as an equivalent of the ownership right in civilian usage. As will be demonstrated later in this chapter, ownership is only a subset of property, one of many property rights in civil law tradition.

Blackstone's view was later rejected by most common law scholars and the judiciary, who embraced the 'bundle of rights' concept, as discussed later. His position is best reflected in this famous quote:

> There is nothing which so generally strikes the imagination, and engages the affections of mankind, as the right of property; or that sole and despotic dominion which one man claims and exercises over the external things of the world, in total exclusion of the right of any other individual in the universe. And yet there are very few, that will give themselves the trouble to consider the origin and foundation of this right.[8]

While legal scholars have mainly expressed the views mentioned in brief above, the usage of the term and its definition become even more diverse if we look at other disciplines and contexts. Thus, for example, Grey tried to summarise contemporary usages of the term 'property' in law, economics and legal theory.[9] One of the usages amongst teachers and law students in England is 'the whole body of law concerned with the use of land'; another is inherent to lawyers and some economists, who identify it with rights *in rem* (rights against the whole world), as opposed to rights *in personam* (rights against a determinate person).

Economists, such as Posner and Demsetz, use the word to indicate the purpose of property, including all rights with the purpose to advance allocative efficiencies, such as the rights to life, liberty and personal security. Others invoke the 'new property', like Reich, arguing that the traditional purpose of

7 Ibid. 110.

8 SW Blackstone, *Commentaries on the Laws of England (1765–1769)* (18th edn, S. Sweet etc. 1829) Book II, chapter 1.

9 TC Grey, 'The disintegration of property' in Pennock and Chapman (n 2).

property is to ensure independence and security, thus proposing a revision of the concept in terms of welfare and public education law.[10]

Another specialised usage defines property as opposed to the liability according to the remedies available to protect it. Thus, property can be enforced, amongst other options (e.g. vindication, restitution), both by injunction and by criminal law sanctions, whereas obligations are usually followed with damages as a remedy.

In Grey's elaboration of property, the author concludes that 'from a glance at the range of current usages, the specialists who design and manipulate the legal structures of the advanced capitalist economies could easily do it without using the term property'.[11]

From a philosophical point of view, Waldron argues that property is 'a concept of which many different conceptions are possible',[12] and the conceptions are relative to different societies and their respective conception of incidents of ownership.[13] Further, defining the general concept, he states it is 'a system of rules governing access to and control of material resources'.[14] Resource further is 'a material object capable of satisfying some human need or want'.[15]

Now that we have sketched the notion of property and possible viewpoints and stances regarding its definition, the concept itself should become more apparent and familiar after identifying its main features and categories, focusing on the common law conceptions. Also, the definitions and features will reflect the currently predominant capitalist economy conceptions of property in the Western world, notwithstanding different conceptions elsewhere, which reflect cultural, economic and societal characteristics of these societies (e.g. socialist conceptions and rejection of private property, tribal and indigenous conceptions of property). The reason for focusing on the former is that this book is primarily looking for practical and policy solutions in the current socio-economic system in the UK and US. Therefore, property in its Western conception is used as a tool to suggest solutions for the transmission of digital assets on death, in the system 'as is'.

10 C Reich, 'The new property' (1964) 73 *Yale Law Journal* 733, discussing the wealth allocation of property function in relation to 'government largesse', different social benefits and services. For an interesting overview, see Harris (n 3) 149–51.

11 Grey (n 9) 71–3.

12 J Waldron, *The Right to Private Property* (Clarendon Press 1990) 31.

13 Ibid.

14 Ibid.

15 Ibid.

(a) Incidents of Property

We will identify critical incidents of property and use them further in the discussions on whether digital assets include these incidents to constitute property.

Honoré has offered one of the best-known attempts at defining the common elements or features of property or ownership. He identifies eleven exhaustive incidents of ownership that 'are not individually necessary, though they may be together sufficient, conditions for the person of inherence to be designated as "owner" of a particular thing in a given system'.[16] These incidents are the right to possess, the right to use, the right to manage, the right to the income of the thing, the right to the capital, the right to security, the rights of transmissibility and absence of term, the prohibition of harmful use, the liability of execution, and the incident of residuarity.[17]

Interestingly, these incidents do not explicitly include the right to destroy and exclude, though these could arguably fall under the rights to manage or right to use and the prohibition of harmful use. Thus, it looks like the quality of absoluteness discussed below and pertaining to civilian systems has not been considered as primary in this theory. However, since it strives to discover common incidents for different legal regimes, this omission is entirely understandable.

Similarly, building on Honoré's theory of property and his eleven elements of the notion, Becker proposes thirteen elements, namely the right (claim) to possess, the right (liberty) to use, the right (power) to manage, the right (claim) to the income, the right (liberty) to consume and destroy, the right (liberty) to modify, the right (power) to alienate, the right (power) to transmit, the right (claim) to security, the absence of term, the prohibition of harmful use, liability to execution, and residuary rules.[18] He further argues that a person who has any one of the first eight elements plus the right to security then has a property right and there are 4,080 such combinations possible (mathematically calculated). Like Honoré, Becker argues that full ownership consists of all the elements in the same individual.[19]

Conversely, unlike Honoré, Becker considers the right to destroy as one of the incidents of ownership. Furthermore, his theory focuses on property as a relative concept, consisting of different interests, with ownership as an absolute one. This concept of a 'bundle of rights' is a dominant feature of the

[16] Honoré (n 5) 112–13.
[17] Ibid. 113–28.
[18] L Becker, 'The moral basis of property rights' in Pennock and Chapman (n 2) 190–1.
[19] Ibid. 192.

common law tradition and will be used in the case studies chapters as a starting point for analysing the legal nature of specific digital assets.

The bundle of rights theory permeates both American and English legal thought and practice, and it is adopted as a practical and flexible approach in our attempt to conceptualise digital assets. This theory is based on Hohfeld's theory of legal relations as jural opposites (rights–no-rights; privilege–duty; power–disability; immunity–liability) and jural correlatives (right–duty; privilege–no-right; power–liability; immunity–disability), which define legal concepts, including property, as a set of these correlatives and relations.[20]

Penner notes that it is 'clearly dominant in the United States, where even the Restatement of Property begins with a Hohfeldian outline of rights and duties, and where the "bundle of rights" is regularly cited by courts in important property cases',[21] and that it is undoubtedly prevalent in England as well. Lawson and Rudden, for example, refer to the law of property as providing an owner with 'a bag of tools'.[22] For Penner, this indicates that property 'is a concept without a definable "essence"; different combinations of the bundle in different circumstances may all count as "property" and no particular right or set of rights in the bundle is determinative'.[23] In this regard, he refers to Munzer's characterisation:

> The idea of property – or, if you prefer, the sophisticated or legal conception of property – involves a constellation of Hohfeldian elements, correlatives, and opposites; a specification of standard incidents of ownership and other related but less powerful interests; and a catalogue of 'things' (tangible and intangible) that are the subjects of these incidents. Hohfeld's conceptions are normative modalities. In the more specific form of Honoré's incidents, these are the relations that constitute property.[24]

Thus, the bundle of rights theory could be best presented and understood as a Hohfeldian–Honorian bundle of jural correlatives, opposites and incidents. This theory will be referred to in the case study chapters later in this book.

Additionally, discussions on conceiving digital assets as property will also refer to significant economic features of property objects. These qualities,

[20] WN Hohfeld, 'Some fundamental legal conceptions as applied in judicial reasoning' (1913) 23 *Yale Law Journal* 16.

[21] *Moore* v *Regents of the Univ of Cal*, 793 P.2d 479 (Cal 1990); *Keystone Bituminous Coal Assn et al.* v *DeBenedictis, Secretary, Pennsylvania Department of Environmental Resources, et al.*, 480 US 470 (1986).

[22] JE Penner, 'The "bundle of rights" picture of property' (1996) 43 *UCLA Law Review* 43, 713.

[23] Ibid. 724.

[24] Ibid. 724.

arguably, qualify various objects as property. The list presented here is not exhaustive; instead, only the most crucial features will be defined here, namely rivalrousness, excludability, persistence (temporality) and interconnectivity.

Rivalrousness (subtractability) is an economic term primarily relating to consumption and the physical quality of an object. It arises in situations 'where one person's use subtracts from the available benefits for others'.[25] In other words, it means that consumption cannot be common for these resources.[26] Most tangible objects are intrinsically rivalrous (e.g. food, clothing, most private goods), whereas most intangible ones are non-rivalrous (information, objects of IP, but not domain names, radio spectrum). Related to this concept is a notion of excludability, meaning the individual's power to control the use of an object.[27] It usually depends on the property rights granted to enable a person to exclude others from using an object.[28] The two terms should not be confused since, for instance, non-rivalrous objects can be excludable. An obvious example is IP, mainly non-rivalrous but excludable since the law has granted exclusive protection, a monopoly, to it. Here the valid question is the cost of excludability and not the exclusion itself.

Persistence is another quality of property objects, both tangible and intangible. It does not mean permanence *per se*; it only implies a certain degree of stability.[29] On the other hand, theoretically, property rights have an indefinite duration, unlike IP, whose duration is still time-limited, conferring different terms of protection to certain kinds of IP.

Interconnectivity, another characteristic of objects in the real world, means that they can affect each other 'by the laws of physics';[30] they are connected and can be perceived as such by senses, or more than two people can experience the same property at the same time.

2.2 Key Categories of Property in Common and Civil Law Countries

After having shown the possible ways to define property and the primary features of the concept, we will further demonstrate the main differences at conceptual and practical levels of consideration of property in two major law traditions, namely common and civil law. Only the most representative

[25] C Hess and E Ostrom, *Understanding Knowledge as a Commons from Theory to Practice* (MIT Press 2007) 9, 352.

[26] DL Weimer and AR Vining, *Policy Analysis* (5th edn, Longman 2011) 72.

[27] Ibid.

[28] Ibid.

[29] See TJ Westbrook, 'Owned: Finding a place for virtual world property rights' (2006) *Michigan State Law Review* 779, 782, 783.

[30] J Fairfield, 'Virtual property' (2005) 85 *Boston University Law Review* 1047.

examples will be mentioned, in an endeavour to help understand the categorisation presented afterwards, find common themes, and indicate some solutions for defining and categorising virtual assets and property. The chapter thus assesses which legal system would be more susceptible to the inclusion of new objects of property. The analysis of this distinction will be especially helpful in Chapter 5, where a hybrid common–civil law solution will be introduced for the games and virtual worlds case study.

The differences between conceptions of property in the two prominent legal families could be explained through the historically and culturally conditioned viewpoint. Continental legal scholars prefer more clearly and coherently defined concepts and theories, almost a dogmatic approach. Thus, Bouckaert notes, 'continental legal science puts a much stronger emphasis on definitions and general principles than its Anglo-American counterpart. Continental jurists at one time identified this conceptual level of legal science as "legal dogmatic" or "legal theory."'[31] Moreover, as Penner remarks, in the common law, 'The specialist fragments the robust unitary conception of ownership into a more shadowy "bundle of rights".' As a result, he claims, the law and legal theorists in common law systems no longer have one single, coherent concept of property.[32]

Nevertheless, the differences and possible similarities between different property concepts have not been exciting to comparative law scholars. As van Erp notes, comparative law focuses on the law of obligations because of the dominant elements of convergence, while property law has not been an area of great interest for comparative lawyers.[33] Furthermore, he argues that the comparative lawyers have taken for granted fundamental, historically rooted differences between civil and common law, considering that there is no possibility for any convergence.[34] Van Erp calls this approach to property law 'technocratic conservatism', 'a legal mentality that aims at preserving the status quo and that accepts changes only when these are completely unavoidable'.[35] However, he predicts shifts in this mentality due to global market integration. He also notes that property law has never been as static as generally considered, giving examples of influences of case law in Germany and the Netherlands (transfer of ownership for security purposes), on the one

[31] B Bouckaert, 'What is property?' (1990) 13 *Harvard Journal of Law and Public Policy* 775, 775–6.
[32] Penner, '"Bundle of rights"' (n 22) 769.
[33] See S van Erp, 'Comparative property law' in M Reimann and R Zimmermann (eds), *The Oxford Handbook of Comparative Law* (Oxford University Press 2006) 1044.
[34] Ibid. 1044–5.
[35] Ibid. 1048.

hand, and changes in common law affected by statutes (Law of Property Act 1925), on the other.[36] The evidence that this view is sound is the attempts at unifying private law at the EU level (see the discussion below). Concerning van Erp's prediction, it is suggested that the global nature of the Internet and the increasing importance of digital assets will contribute to a change of mentality in the long term.

When discussing the difference in concepts of property between common and civil law systems, it is essential to note that within the EU, property is still subject to the national law of member states.[37] Thus, the distinction between common and civil law conceptions is still in place, even if we disregard the US conception, which inherited and received the common law tradition, and currently mainly stands by the aforementioned bundle of rights theory.

There are significant moves within the EU to harmonise aspects of national property laws, provided that we accept the argument that IP is property.[38] Notably, Article 118 of the Treaty on the Functioning of the European Union (Lisbon Treaty) calls for the 'creation of European intellectual property rights to provide uniform protection of intellectual property rights throughout the Union'.[39] There have been similar global attempts to harmonise IP law, within WIPO, WTO, multilaterally and bilaterally, by different treaties and conventions.[40] Though this aim is currently far from being achieved, it sheds important light on the possible intentions of unifying IP.

[36] Ibid. 1048–9. Here he argues that the Law has introduced the *numerus clausus* principle in common law, limiting the number of estates in s 1.

[37] Article 345 (ex Article 295 TEC) of the Treaty on the Functioning of the European Union OJ C 83/47 30/03/2010.

[38] The debate over whether IP is, in effect, property or a statutory monopoly and privilege is outside the scope of this book. For more on the rich academic debate, see for example J Hughes, 'Copyright and incomplete historiographies: Of piracy, propertisation, and Thomas Jefferson' (2005–6) 79 *Southern California Law Review* 993; M Kretschmer, L Bently and R Deazley, 'The history of copyright history (revisited)' (2013) 5(1) *WIPO Journal* 35; L Lessig, *Free Culture: How Big Media Uses Technology and the Law to Lock Down Culture and Control Creativity* (Penguin 2004); MA Lemley and PJ Weiser, 'Should property or liability rules govern information?' (2007) 85 *Texas Law Review* 783; HL MacQueen, 'The war of the booksellers: Natural law, equity, and literary property in eighteenth-century Scotland' (2014) 35(3) *Journal of Legal History* 231; HL MacQueen, 'Intellectual property and the common law in Scotland c1700–c1850' in CW Ng, L Bently and GD Agostino (eds), *The Common Law of Intellectual Property: Essays in Honour of Professor David Vaver* (Hart 2010). In addition, an interesting view of IP being an example of the Scots law concept of 'exclusive privilege' has been offered by Black. See G Black, *A Right of Publicity in Scots Law* (PhD thesis, University of Edinburgh 2009) 192–4, 205–10.

[39] Treaty on the Functioning of the European Union (n 37).

[40] The most important international IP guidelines involve four treaties: the Paris Convention for the Protection of Industrial Property of 20 March 1883 (as revised); the Madrid Agreement

Furthermore, there are efforts to unify contract law or create some legal core that would foster the Single Market and benefit consumers at the EU level. The proposals range from a non-binding 'toolbox' that would improve consistency and harmonise member states' contract rules to an all-encompassing contract or even civil code for the EU.[41] The process has gone as far as a proposal for a common framework for sales law,[42] which means that in the short to medium term, we may expect a European code of some kind, contract or civil, or even the harmonising of private law in general.

Differences between the legal families are notable if we consider property remedies. In common law, the primary remedy is damages and not the return of the property like in civil law. Thus the law of tort (trespass and conversion) deals with this protection rather than the law of property.[43] Discussing remedies in English law, Reid notes that the remedies for recovering the possession of the thing owned are provided by the law of obligations, based on tort. However, even then, as he interestingly puts it, 'the law of obligations is the servant and not the master' because in order to use the remedy provided by the law of obligations, one must first own the property.[44]

One of the most relevant classifications of property objects for our discussion is that of tangible and intangible property. Digital assets are inherently intangible, so this analysis will be relevant to each of the case studies.

While common law recognises various items and phenomena as things, making them susceptible to the property law concept, the same cannot be said of civil law. The German Civil Code (BGB) restricts property to things, defined therein as tangible and corporeal.[45] Consequently, an incorporeal

Concerning the International Registration of Marks of 14 April 1891 (as revised); the Berne Convention for the Protection of Literary and Artistic Works of 9 September 1886 (as revised); and the Trade-Related Aspects of Intellectual Property Rights (TRIPS). Also relevant are the WIPO Copyright Treaty (WCT) and the WIPO Performance and Phonogram Treaty (WPPT).

41 See European Commission, 'Green paper from the Commission on policy options for progress towards a European Contract Law for consumers and businesses' COM(2010)348 final. For more, see R Zimmermann, 'Comparative law and the Europeanization of private law' in Reimann and Zimmermann (n 33) 540–77.

42 European Commission, 'Proposal for a Regulation of the European Parliament and of the Council on a Common European Sales Law' COM(2011) 635 final 2011/0284(COD).

43 M Bridge, *Personal Property Law* (3rd edn, Oxford University Press 2002) 47.

44 KGC Reid, 'Obligations and property: Exploring the border' (1997) *Acta Juridica* 225, 226.

45 For more, see EJ Cohn, *Manual of German Law* (British Institute of International and Comparative Law; Oceana Publications 1968) 174–80.

thing does not exist in German private law and cannot be subject to the concept of ownership. This aim has been achieved through other legislation.[46]

The protection of property offered by the basic law of the Federal Republic of Germany 1949,[47] Article 14, was extended by the Federal Constitutional Court,[48] expanding the interpretation of this clause beyond the limits of the private law concept, for the meaning and context of the whole Constitution. This created different property concepts in public, constitutional, criminal and private law, as defined in the BGB. Public property law, created administratively and constitutionally, violated some of the principles of the BGB, such as that property must be corporeal. In this way, electricity was recognised as property.[49] Furthermore, the constitutional property concept included IP rights, a claim for unemployment benefits, and bills of exchange.[50] This phenomenon is known as the theory of dualism of private and public property in German law.[51]

In French law, the word *biens* is used to designate both tangible and intangible property. The term is used in the Code Civil instead of *chose* meaning thing, to designate a value rather than a thing itself.[52] Incorporeal things are considered movable property by prescription of law, but this category is not as broad as that of English common law (Article 527 of the French Civil Code).

On the other hand, English common law has been quite flexible when applying the concept of ownership to intangibles, using the same principle as for tangibles. Thus, Maitland argues that 'any right or group of rights that is of a permanent kind can be thought of as a thing . . . mediaeval law is rich with incorporeal things'.[53] As Lawson and Rudden note, this approach is still

[46] Trademarks: Markengesetz of 25.10.1994. BGBI. I S. 1273; Patents: Patentgesetz of 16.12.1980. BGBI. 1981 I S. 1; Copyright: Urheberrechtgesetz of 9.91965. BGBI. I S. 1273. See generally MJ Raff, *Private Property and Environmental Responsibility: A Comparative Study of German Real Property Law* (Kluwer Law International 2003) 191–3; W Ebke and MW Finkin, *Introduction to German Law* (Kluwer Law International 1996) 227.

[47] See AJ van der Walt, *Constitutional Property Clauses: A Comparative Analysis* (Kluwer Law International 1999) 121–63.

[48] BVerfGE 51, 193 (Warenzeichen case) 1979 219 in van der Walt (n 47) 151.

[49] See Raff (n 46) 164–5. See similarly Ebke and Finkin (n 46) 72, 73.

[50] See the leading case describing the constitutional guarantee of property: *Hamburg Dyke Case* (1969) 22 NJW 309 and Raff (n 46) 170–1.

[51] Raff (n 46) 164.

[52] J Bell et al., *Principles of French Law* (2nd edn, Oxford University Press 2008) 270.

[53] Quoted in FH Lawson and B Rudden, *The Law of Property* (Oxford University Press 2002) 82.

being used in the cases of 'owner of patent' or a mortgage.[54] Discussing rights as things, Reid argues that incorporeal assets and rights are already a part of a person's patrimony, and their importance has increased significantly. Thus, he concludes, 'It seems odd to deny them the status of things.'[55] However, as seen in the case of information later in this book, this does not necessarily have to be true. English law is still very reluctant to recognise property in information, as seen in the example of emails in Chapter 6. Some recent developments suggest this may change in the future, especially concerning crypto assets and cryptocurrencies.[56] However, at the time of writing, the law reform is still at a very early stage, and the uncertainty remains.

Prima facie, it seems that intangibility will not represent an obstacle to recognising digital assets as property, especially in English law. However, this proposition does not prove true in the case of information and personal data, where the analysis in Chapter 6 demonstrates that only US law occasionally shows readiness to recognise certain types of information as property. This inconsistency serves as another argument against recognising digital assets as property.

Another peculiarity of common law, when compared with civil law systems, is the category of choses in equity and equitable ownership. This concept will be analysed to show some successful examples of legal transplants and cross-pollination between legal systems, providing support for hybrid solutions offered in Chapter 5 for games and virtual worlds.

Historically, English equity law was created by the Chancellor and the Court of Chancery, and unified by statute at the end of nineteenth century to make it applicable to one system of courts. However, it is unclear whether these two systems fused, and this question for Penner 'remains controversial'.[57]

The most significant contribution of equity is the trust, existing when a fund of property is held by a legal owner, a trustee, with an enforceable legal obligation that it be used for another, the beneficiary.[58] Alternatively, simply put, it is 'a flexible grouping of people and property in which one group of people (trustees), look after assets for another group (beneficiaries)'.[59] Trustees are referred to as 'the legal owners', because the property is in their

[54] Ibid.

[55] Reid (n 44) 231.

[56] Law Commission, 'Digital assets: Call for evidence' (April 2021) <https://www.lawcom.gov.uk/project/digital-assets/> last accessed 1 August 2021.

[57] Penner, '"Bundle of rights"' (n 22) 1332.

[58] For more and a comparison with choses in action see J Penner, *The Idea of Property in Law* (Oxford University Press 1997) 133–8.

[59] J Ball, 'The boundaries of property rights in English law', Report to the XVIIth International Congress of Comparative Law (2006) 10(3) *Electronic Journal of Comparative Law* 8.

name, but they do not enjoy the benefit of the property. They simply manage the trust for the benefit of the real, equitable owners – the beneficiaries. Trust is usually formed by contract, but this is not a rule. In the case of implied or constructive trust in England, no formality is needed to establish a trust,[60] apart from the constructive trust arising when a person deals with property in an 'unconscionable manner'.[61] Another type of constructive trust arises in the sale contracts, from the moment the price is paid, where it is deemed that a property is held in constructive trust for a transferee.[62]

The nature of trust, however, is still to be resolved. While proponents of Scott's view argue that it is a branch of property law, Langbein considers trust a contract, highlighting that the power and duties of the trustee are the default contract terms between the settlor of the trust and the trustee.[63]

Merrill and Smith identify characteristics of both regimes in trust, especially given the complicated position of the trustee who is subject to both property and contractual regimes. They put it succinctly: 'Thus, we can think of a trust as the transfer of in rem rights associated with ownership, subject to a set of in personam duties designed to fulfil the settlor's intentions toward the beneficiary.'[64]

Elements of both regimes, and particularly on the side of beneficiaries, are detected by Ball, too. She notes that while a beneficiary's rights in relation to a trustee are personal, the beneficiary also has real, property-like relations to third persons.[65]

Hansmann and Mattei are amongst those who argue that the principal contributions of trust are those related to property-like aspects. Here they refer to an interesting feature of property, explaining, 'When we say that assets are someone's property, we generally mean (among other things) that those assets are presumed available to satisfy claims of that person's creditors.'[66]

The most important contribution of trust law, in Hansmann and Mattei's opinion, is that of arranging relations between the trust parties and third parties, including creditors, relations that cannot be easily dealt with by the rules of contract and agency law in civil law countries, at least not without significant transaction costs. The critical feature of a trust is 'partition of a

[60] See Law of Property Act 1925, s 53(2).

[61] See Millet LJ in *Paragon Finance plc* v *DB Thakerar & Co* [1999] 1 All ER 400.

[62] See *Chinn* v *Collins* [1981] AC 533.

[63] As set out in H Hansmann and U Mattei, 'The functions of trust law: A comparative legal and economic analysis' (1998) 73 *New York University Law Review* 434, 469–70.

[64] TW Merrill and HE Smith, 'The property/contract interface' (2001) 101(4) *Columbia Law Review* 773, 844.

[65] Ball (n 59) 9.

[66] Hansmann and Mattei (n 63) 470.

discrete set of assets', managed separately and capable of being security to creditors, as a distinct entity.[67]

Unlike the Anglo-American tradition, where a dual system of common law and equity arose, the latter creating the concept of trust, property law in civil law countries was predominantly developed by academic lawyers, based on the Roman law tradition. The central concept was the obligation, which framed trust-like arrangements,[68] like the Roman concept of *fiducia*, creating contractual, not property, relationships between parties.[69] However, trust as property, not obligation, would be contrary to the civil law doctrine of the unity of property, the *numerus clausus* principle mentioned later. This doctrine was developed after the French Revolution, regarding the division of property rights as a relic of feudalism.[70]

Nevertheless, an institution resembling trust can now be found in French law. That is *fiducie*, a trust-like institution, introduced in the French Civil Code by the Law of 19 February 2007, amending former Articles 2011 to 2031 of the Civil Code. Although the definition from Article 2011 is similar to the common law concept,[71] it is only applied to relations *inter vivos*, which is not the case in common law. Furthermore, it must be made clear, it is a formal contract and there is nothing like the implied or constructive trust in England discussed earlier in this section. Also, *fiducie* is not open to individuals but only to institutions. As discussed above, its nature is contractual, whereas English trust has a mixed nature, involving both contractual and property elements. Finally, and importantly, the most innovative aspect of this institute is that it is separate from the fiduciary's personal assets, which breaks the unitary conception of *patrimoine* (estate, all rights *in rem* and *in personam*)[72] in French law.[73]

This institute is an example of a legal transplant between common and civilian legal systems.[74] It shows once more that a useful legal concept, effective in a legal and economic milieu, can be exported and recognised in a different legal system that, at first glance, would not be susceptible to its reception. One of the reasons for introducing this concept is related to globalisation and the fact that companies were moving their assets to jurisdictions with

67 Ibid. 466.
68 Ibid. 441.
69 Ibid. 443.
70 Ibid. 442.
71 Article 2011 du Code Civil (loi du 19 février 2007).
72 E Steiner, *French Law: A Comparative Approach* (Oxford University Press 2010) 379.
73 For more, see ibid. 387–9.
74 A Watson, *Legal Transplants: An Approach to Comparative Law* (Scottish Academic Press 1974) 20.

more flexible and competitive legal instruments.[75] It confirms Watson's argument that transplanting is the most fertile source of development. Changes in many legal systems are the results of borrowing. According to Watson, the law is similar to technology, it is 'the fruit of human experience', and when a rule is invented by one nation, it can be appreciated and used for the needs of many other people.[76] However, since *fiducie* is not an exact or close replica of trust, with essential features missing (e.g. post-mortem features), it cannot be seen as a very representative example either, at least not yet.[77] Nevertheless, these law reforms have introduced a previously unimaginable concept into a civil law legal system, demonstrating the potential for cross-pollination between legal systems and providing an argument for reform proposals suggested in this book.

(a) Some Consideration of Categories of Property in Common Law

The historical development of the property concept in common law systems is characterised by the progressive relativisation of this, initially, absolutely defined concept and the dephysicalisation of the notion. This development has been soundly described and discussed by Vandevelde.[78] He describes the shift from Sir William Blackstone's eighteenth-century conception to the nineteenth century.[79] The exceptions to the physicalist and the absolutist elements of Blackstone's conception were incorporated into the law by courts, which sought to protect beneficial interests as property 'even though no thing was involved'. Vandevelde sees the rationale for that in a theory of natural law and in the instrumentalist public policy of the state. Vandevelde calls this phenomenon, the protection of value rather than things, 'the dephysicalisation of property'.[80]

Further developments led to the complete abandonment of the Blackstonian conception of property by the beginning of the twentieth century. The most outstanding contribution to the establishment and formulation of a new conception was made by Hohfeld, in his theory discussed earlier.[81] Property rights were no longer seen as absolute but limited, without

75 Steiner (n 72) 388. See also Watson (n 74) 97.

76 Watson (n 74) 95, 100.

77 See for example D Baudouin, 'Fiducie in French law' (2007) 2 *International Business Law Journal* 276, 276–81; P Matthews, 'The French fiducie: And now for something completely different?' (2007) 21(1) *Trust Law International* 17, 17–42.

78 KJ Vandevelde, 'The new property of the nineteenth century: The development of the modern concept of property' (1980) *Buffalo Law Review* 29, 325.

79 Ibid. 330. See also Section 1.1.

80 Vandevelde (n 78) 329.

81 Hohfeld (n 20).

favouring any set of rights over others. As Vandevelde usefully summarises, 'the particular combination of rights that comprised property in a given case would be decided according to the circumstances'.[82]

These developments have not been followed in the civil law family, where the concept of absolute ownership and the *numerus clausus* principle (finite number of property rights, prescribed in statutes or codes) are predominant, both in legal literature and statutory rules.[83] The principle of *numerus clausus* in German law means that the BGB limits the number, kind and content of rights *in rem*. The reason underpinning this principle is to enable third parties to assess risks arising from absolute rights.[84]

In England, as McKendrick describes, property rights are predominantly made by judges and have not yet been incorporated into statutes (apart from the Law of Property Act 1925), unlike in civil law countries where property law forms a significant part of civil codes or statutes. However, it is uncertain whether such a practice will continue.[85] However, as he notes, quoting Lord Wilberforce in *National Provincial Bank Ltd* v *Ainsworth*,[86] there is a 'continuing creative ability of the courts' to recognise the right or interest in the category of property if it is 'definable, identifiable by third parties, capable in its nature of assumption by third parties, and have some degree of permanence or stability'.[87] As far as England is concerned, Lawson and Rudden note that 'the English legislator – perhaps wisely – gives merely inclusive definition covering both objects and interests in them'.[88]

Further, property in England historically developed from relations involving land and tangible assets within the framework of common law (except a few statutes like the Law of Property Act).[89] Therefore, it had to evolve from

[82] Vandevelde (n 78) 336.

[83] N Horn et al., *German Private and Commercial Law: An Introduction* (Clarendon Press 1982).

[84] Ebke and Finkin (n 46) 230.

[85] See E McKendrick in N Palmer and E McKendrick (eds), *Interests in Goods* (Lloyd's of London Press 1993) 41.

[86] [1965] AC 1175, 1247–8.

[87] Lord Wilberforce in McKendrick (n 85) 40.

[88] Lawson and Rudden (n 53) 10; Theft Act 1968, c 60; Law of Property Act 1925, c 20; Trustee Act 1925, c 19 (the most comprehensive definition, including real and personal property, estate shares and interests in property, debt, anything in action, any other right and interest whether in possession or not; see s 68(11). However, the Trustee Act 2000, c 29, in s 39(1) completely abandoned the notion of property referring to assets as including 'any right or interest').

[89] 'The logic of property law is bound up with tangibility.' See A Hudson, 'The unbearable lightness of property' in A Hudson (ed.), *New Perspectives on Property Law, Obligations and Restitution* (Cavendish 2004) 9.

precedents developed concerning land, primarily to deal with complex chattels and intangible assets. The consequence of this was that the concept and the logic behind it 'was likely to be stretched too far'.[90] In addition, even the existing legislation did not help shed much light on the concept of property in England since the Law of Property Act 1925 classifies property instead of defining it. Therefore, Ball rightly notes that 'The nature of property rights frequently has to be deduced from a piecemeal collection of *ad hoc* definitions, usually in case law, borrowing from various sources.'[91]

There is a general preference in English law to focus on remedies rather than principles, in that, 'in English law, the legal question tends to be not necessarily based on the definition of the property but whether something can be protected by the law or handled by the law, perhaps on death, by a transfer or in the law of theft'.[92] Thus in common law, there is 'a predominance of procedural law as means for the delivery of justice', whereas, in France, codification from the early 1800s separated procedural law from the legal principles and substantive law. Therefore, if the French case can be brought within the definitions, then the procedural aspect of the case and, disputably, justice would follow.[93] English law, especially the law of equity, prefers procedural principles.[94] As Samuel argues, French law starts from a subjective right (*le droit subjectif*), and if there is a right, a remedy will be found, whereas English law starts from actions, and there has to be an action and a remedy in order to have an enforceable right.[95] The case law confirms this argument.[96]

In English law, property is divided into personal and real property. Real property refers to land, while personal is everything else. Personal property is residual, and, as Bridge argues, the 'somewhat formless nature of the subject' makes it capable of extension, 'both in respect of recognition of novel kinds of property and of its quantity'.[97] Until the mid-nineteenth century, personal property could not be recovered *in rem* (i.e. the property could not be

[90] Ibid.

[91] Ball (n 59) 4.

[92] Ibid. 8.

[93] This is according to the famous principle, accepted in French legal tradition, 'ubi ius, ibi remedium' (when there is a right, there is a remedy, too). See ibid. 8.

[94] Ibid.

[95] G Samuel, '"Le droit subjectif" and English law' (1987) 46(2) *Cambridge Law Journal* 264, 264–86; Bell et al. (n 52) 7.

[96] See for example Sir Nicolas Browne – Wilkinson VC in *Kingdom of Spain* v *Christie, Manson & Woods Ltd* [1986] 1 WLR 1120, 1129; *F* v *Wirral MBC* [1991] 2 WLR 1132 (CA).

[97] Bridge (n 43) 1.

retrieved as such), as was the case for real property and actions *in rem*.[98] Only a claim for monetary damages was allowed until the Common Law Procedure Act 1854 (s 78),[99] which enabled the return of the chattels detained as a primary remedy.[100]

The personal property further comprises chattels real (leasehold interests in land) and chattels personal. Chattels personal are divided into choses in action and choses in possession.[101]

Choses in possession are tangible, movable things (when they are a subject of sale, they are considered goods), and are more regularly referred to as chattels.[102]

Choses in action are a different type of intangible (incorporeal) property, residual as well. As Bridge argues, they pertain to what remains after eliminating the choses in possession (i.e. debts, goodwill, rights under an insurance policy, shares, bills of exchange and IP).[103] The main characteristic of choses in action is that they can only be claimed by action, legal procedure, and not *in rem*, reclaiming possession.[104] Scholars like Hudson label choses in action 'quasi property', a purely personal claim, recognised as property due to its transferability (the transfer pertains only to the right to receive something) and 'separability' (established by Penner and discussed earlier).[105]

Choses in action further divide into pure intangibles (e.g. debt, goodwill, copyright) and documentary intangibles (bill of lading, bill of exchange, promissory note, shares, insurance policy).[106] These categories fall under the group of common law property. In addition, there exists equitable property (i.e. trust), discussed earlier.

98 Note the possible confusion between the real and personal property and rights *in rem* and *in personam*. While the former are concerned only with property and thus defined only in common law systems, the latter concern division of rights, differentiating between the rights against the whole world (*in rem*, property rights) and only against a particular person (*in personam*, contractual rights etc.).

99 Bridge (n 43) 2.

100 Specific Delivery of Chattels, Common Law Procedure Act 1854, c 125, s 78.

101 'All personal things are either in possession or in action, the law knows no tertium quid between the two.' The famous dictum of Fry LJ in *Colonial Bank* v *Whinney* (1885) 30 Ch D 261, a view shared by the House of Lords on appeal: (1886) 11 AC 426.

102 Bridge (n 43) 3.

103 Ibid. 4.

104 *Choses in action* are 'personal rights of property which can only be claimed or enforced by action and not by taking physical possession' or 'a thing which you cannot take, but must go to law to secure'. See *Torkington* v *Magee* [1902] 2 KB 427, 430; TC Williams, 'Property, things in action and copyright' (1895) 11 *Law Quarterly Review* 223, 232.

105 Hudson, 'Unbearable lightness of property' (n 89) 23–30.

106 Bridge (n 43) 6–9.

The example of the California Civil Code illustrates similarities with the English categories of property. The general division has been slightly altered and refers to real or immovable and personal or movable property.[107] The general divisions and effects are still very similar, and the property law regimes share vocabulary and principles.[108] Some differences and exceptions will be emphasised in examples relevant to this book, such as property in information.

Choses in action are an exciting peculiarity of common law, for some authors representing 'compromised forms of property'.[109] As Penner explains, the proprietary character of choses in action shows when 'things go badly wrong'. Penner interestingly explains their 'thinghood': 'It is not because they are alienable that they are "things" of this kind. Rather, they are alienable because they are things.' This means they do not have a quality of thinghood, namely alienability, that would naturally qualify them as things. Instead, they have been recognised by law as things, arguably quite artificially, and the law has attributed alienability to them. The classic example is debt, recognised as property in common law and assigned a feature of alienability, unlike in civilian law. Another example is IP, which does not intrinsically have all the classical property features (e.g. exclusivity, rivalrousness, corporality), but has been recognised as property and, arguably, put into this category.

The importance of finding the common elements and themes in civil and common law conceptions of property will be demonstrated further in Chapter 5, where a solution based on both traditions will be suggested. It is argued in this book that the legal borrowing Watson suggests is even more desirable on the Internet, with its blurred boundaries and jurisdiction issues.

(b) Property Rights vs Obligations

When discussing the concept of property, one should inevitably refer to obligations, especially the law of contracts, as its correlative concept. This is particularly significant for this book, as most of the currently recognised rights of users in their digital assets are contractual (as demonstrated in the following chapters).

Based on Roman law concepts, most authors in common and civil law systems would use the terms '*in rem*' and '*in personam*' when referring to property and obligations. In common law, the term 'obligations' is rare although

[107] California Civil Code, s 657.

[108] See generally DA Thomas, 'Anglo-American land law: Diverging developments from a shared history – part III: British and American real property law and practice – a contemporary comparison' (1999–2000) 34 *Real Property, Probate and Trust Journal* 443.

[109] Penner, *Idea of Property* (n 58) 107.

it has recently come into frequent use by courts and scholars. Instead, common law jurisprudence refers separately to the notions of contracts, torts and unjust enrichment. There has been, as some argue, a reception of obligations as a category in English law where more judges and academics speak of them when addressing one of these three categories.[110]

Civilian lawyers would commonly use the terms 'real' and 'personal rights', a usage that should not be confused with that of real and personal property in common law (both real and personal property belong to *in rem* property rights in common law, the first about land and the second about everything else). Property rights, as rights *in rem*, are rights against the whole world, a potentially infinite number of persons, while contracts, rights *in personam*, according to the doctrine of privity of contract in common law, and statutory rules and theory in civil law, bind only the parties to the contract, not third parties.[111] Thus, rights *in rem* are often referred to as absolute rights, whereas rights *in personam* are relative; their effect pertains only to the contract parties (with some exceptions).[112]

In an attempt to differentiate rights *in rem* from rights *in personam*, Reid suggests three unique features of real rights: real rights concern things; they can be enforced against the whole world; and the obligation correlative to a real right is negative (to refrain from doing something).[113] American authors Merrill and Smith identify 'four differentiating features of in rem rights' that could be subsumed under the three elements identified by Reid.[114] In civilian systems, the border between property and obligations is relatively straightforward. This clarity can be attributed to the *numerus clausus* principle, which states that all real rights are enumerated and defined in codes or statutes, and parties cannot invent new rights or modify existing ones (with the limitations discussed in previous sections).

[110] See GH Samuel and J Rinkes, *Law of Obligations and Legal Remedies* (2nd edn, Cavendish 2001) 252, 254, 262–9.

[111] Bridge (n 43) 26; Samuel and Rinkes (n 110) 364–7; *Barker* v *Stickney* [1919] 1 KB 121, 132; *McGruther* v *Pitcher* [1904] 2 Ch 306; *Taddy* v *Sterious* [1904] 1 Ch 354.

[112] For a civilian perspective, see Ebke and Finkin (n 46) 228, 229. For mixed and common law systems, see P Sutherland and D Johnston, 'Contracts for the benefit of third parties' in R Zimmermann, D Visser and K Reid (eds), *Mixed Systems in Comparative Perspective* (Oxford University Press 2004) 208–39.

[113] Reid (n 44) 227.

[114] '(1) in rem rights apply to a large and indefinite class of dutyholders; (2) in rem rights attach to persons only insofar as they own particular "things" and not otherwise; (3) all persons hold in rem duties to a large and indefinite class of holders of such rights; and (4) in rem duties are always duties of abstention rather than performance.' See Merrill and Smith (n 64) 773–852, 789.

Thus, rights appearing in statutes in civilian systems as real rights (ownership, usufruct, securities and servitudes) are property rights, while others are obligations.[115] In civil law countries, obligations have a residuary nature, consisting of what is left after deducting real rights. In Reid's opinion, there are also categories of rights that are hard to classify, bearing in mind, on the one hand, their affinity with both real and personal rights (lease), and, on the other, their affinity with neither (trusts or IP). He suggests that while it is possible to attempt classification using the categories of real and personal rights for a trust, classification is impossible for IP and thus is mainly abandoned by authors.[116]

From the economic perspective, discussing the issue of costs concerning property rights and obligations, Merrill and Smith argue that 'information costs are key to understanding the features of a system of property rights'.[117] The advantage of property rights is that they enable low-cost identification and definition of resources, in the case where there is a large number of potential claimants to resources; thus, these *in rem* rights are governed by 'bright-line rules that allow large and indefinite numbers of people to identify owned resources at low cost'.[118] Obligations, on the other hand, are regulated by more flexible rules 'that minimise the costs of tailoring rights and obligations to each particular situation'.[119]

Crucially, property differs from contracts in that it is always transmissible to heirs on the owner's death. All property, personal and real, corporeal and incorporeal, to name but a few kinds, forms inheritance, the estate of a person, and transmits by the rules of succession to the deceased's heirs. In the UK, for instance, 'a person's estate is the aggregate of all the property to which he is beneficially entitled'.[120] In the US, 'probate assets are those assets of the decedent, includible in the gross estate under IRC § 2033, that were held in his or her name at the time of death'.[121]

115 Reid (n 44) 228.
116 Ibid. 229.
117 Merrill and Smith (n 64) 833.
118 Ibid. 793.
119 Ibid. 798.
120 Wills Act 1837, c 26, s 3 (this Act does not extend its effect to Scotland); Inheritance Tax Act 1984, c 51, s 5(1), applicable to England, Scotland, Wales and Northern Ireland. similarly Succession (Scotland) Act 1964, c 41, s 32 defines estate as property belonging to the deceased at the time of death.
121 J Darrow and G Ferrera, 'Who owns a decedent's e-mails: Inheritable probate assets or property of the network?' (2006) 10 *New York University Journal of Legislation and Public Policy* 281. See also Sherrin et al., quoted in L McKinnon, 'Planning for the succession of digital assets' (2011) 27(4) *Computer Law and Security Review* 362, 362–7.

Conversely, obligations in principle do not persist on demise. In common law jurisdictions, purely personal obligations 'die with a person'.[122] This position, however, has been revised, and most of the personal rights of action arising from torts survive on death, according to the Law Reform (Miscellaneous Provisions) Act 1934. The only one that does not persist is defamation. As for contractual rights, personal contracts (e.g. employment contracts, contracts between an artist and a person commissioning them) will be discharged on death unless there is an opposite provision in the contract.[123] A contract that is not personal in nature is not discharged upon death, no matter whether it was broken or not; according to the reform mentioned above, it will survive for the benefit of the estate.[124] A personal representative stands in the position of the deceased without being a party to the contract.[125] In French law, with the notion of *patrimoine*,[126] comprising a person's rights and liabilities, all rights and liabilities pass on to the heir(s), who stand in the position of the deceased.[127] Like English common law, in French law, strictly personal contracts (either by the parties' agreement[128] or the nature of the contract[129]) end on death.[130] The same is true in Germany.[131]

Despite the differences presented, it has been noticed that there is a tendency to weaken the centrality of property rights to the advantage of contracts, torts or trusts in England. This tendency goes as far as to call for awarding

[122] Principle '*Actio personalis moritur cum persona*' in *Beker* v *Bolton* (1808) 1 Camp 439; 170 ER 1033, but the Law Reform (Miscellaneous Provisions) Act 1934, c 41 (as amended), revised the rule mandating that all personal rights will survive against and for the benefit of the estate, with the only exception of defamation and claim for bereavement. For the comparison between the US and German perspectives, see for example H Rosler, 'Dignitarian posthumous personality rights – an analysis of US and German constitutional and tort law' (2008) 26 *Berkeley Journal of International Law* 153.

[123] See *Farrow* v *Wilson* (1869) LR 4 CP 744, 746.

[124] See Law Reform (Miscellaneous Provisions) Act 1934, s 1(1); *Sugden* v *Sugden* [1957] 1 All ER 300.

[125] *Beswick* v *Beswick* (1966) Ch 538. For commentary about the contracts and succession, see AR Mellows, *The Law of Succession* (4th edn, Butterworths 1983) 295–6.

[126] B Nicholas, *The French Law of Contract* (2nd edn, Clarendon Press; Oxford University Press 1992) 29, 30.

[127] For more, see ML Levillard, 'France' in DJ Hayton (ed.), *European Succession Laws* (2nd edn, Jordans 2002) 219.

[128] Code civil, Article 1122.

[129] Ibid. Articles 1795, 2003.

[130] For more, see B Nicholas (n 126) 172.

[131] K Kuhne et al., 'Germany' in B Nicholas (n 126) 244, 257.

equal protection to personal rights and property rights.[132] A prominent example of this is the statement of Lord Nicholls in *Attorney-General* v *Blake*:

> Property rights are superior to contractual rights in that, unlike contractual rights, property rights may survive against an indefinite class of persons. However, it is not easy to see why, as between the parties to a contract, a violation of a party's contractual rights should attract a lesser degree of remedy than a violation of his property rights . . . it is not clear why it should be any more permissible to expropriate personal rights than it is permissible to expropriate property rights.[133]

In response to this tendency, some authors suggest an even more radical solution, proposing a unified civil remedy for property, contracts and trust, invoking the approximation of the main categories of private law.[134] To support this, they point to an increasing willingness of courts to use 'the full armoury of remedies' in English common law.[135] For instance, in the recent case of *Manchester Airport plc* v *Dutton*, the proprietary remedy was granted to the non-owner. The company hired by the owner to conduct clearing works on the land requested repossession (ejectment) from the demonstrators who occupied the land. The demonstrators claimed that the company was not entitled to this remedy since it did not have a title of possession. However, a majority of the Court of Appeal held that the company had a contractual right and was entitled to this remedy. The situation would look very different in the civil law systems, where a more precise division between *in rem* and *in personam* rights and remedies exists, and thus only the owner could claim an *in rem* remedy.[136] Some argue that, because English law has never actually developed proprietary remedies, that is, restitution and vindication, and uses torts to protect property (trespass, nuisance and conversion), it does not have such a strict division between real and personal rights, property and obligations.[137] Recognising the unclear picture regarding remedies applied, Samuel argues that perhaps the notion of obligations cannot be imported

132 See D Pearce, 'Property and contract: where are we?' in Hudson, *New Perspectives on Property Law* (n 89) 109.

133 [2001] 1 AC 268, 283.

134 See Pearce (n 132) 109.

135 Ibid. 116. Another example is *Manchester Airport plc* v *Dutton* [2000] QB 133, where the court allowed a person to use a proprietary remedy to repossess land belonging to another only on the basis of a contractual relationship.

136 For a commentary, see G Samuel, *Understanding Contractual and Tortious Obligations* (Law Matters 2005) 110, 111.

137 Ibid. 6.

into English law, which has different reasoning in respect to the law of things than Roman or Continental law.[138]

Others, conversely, argue that the two legal systems in Europe could be brought closer. Van Erp, for example, proposes a more flexible *numerus clausus*, '*quasi-numerus clausus*', where parties have more freedom to shape property rights and are given the freedom to create new property rights under certain strict conditions. In his opinion, these rights could be effective against certain interested third parties, but not the whole world. This result, according to him, would make property law a 'borderline law, a legal area in which traditional property law is further developed through contract and tort law'. Even more radically, van Erp asserts that the trend towards relaxation and flexibility of property rights 'is a *conditio sine qua non* for the development of property law in an era characterised by regional and global economic integration, with its resulting osmosis between national, European and global property law'.[139] This trend would also help settle the legal nature of the new types of assets, such as digital assets.

2.3 Theoretical Justifications for Property

A theory of property could be conceived in many ways. It could be defined as an attempt at a normative justification for allocating property rights at all or in a particular way (detecting which human interests are relevant to a particular way of allocation); it could provide reasons for allocating resources in a certain way; or it could specify the content of property rights at various levels of generality.[140] Similarly, Becker identifies three levels of property justifications: *general* – whether there should be property rights at all; *specific* – why there should be a specific sort of property right; and *particular* – why a particular person ought to have a particular property right in a particular thing.[141] The discussion in this book will mainly relate to the specific level of justifications. In other words, it will mainly attempt to use normative theories to ascertain whether property in digital assets can be asserted in particular cases, that is, emails, social network profiles, and so on.

The justification for property as a basis for normative and social action can be traced back to the works of most of the great philosophers and social

[138] Ibid.
[139] Van Erp (n 33) 21, 22.
[140] See GS Alexander and EM Peñalver, *An Introduction to Property Theory* (Cambridge University Press 2012) 6.
[141] LC Becker, *Property Rights: Philosophic Foundations* (Routledge and Kegan Paul 1977) 23.

thinkers, from the great Greek authors (Plato and Aristotle),[142] to Roman doctrine and law,[143] to natural law scholars, utilitarians, liberals and socialists, to name but a few.

It is important to note at the outset that, although these theories will not be elaborated further here, key Greek property concepts were used as a basis for later discussion about property and many thinkers used them as a starting point for their theories. Thus, the Greeks developed the idea of natural law, which evolved during the Roman period[144] and advanced throughout the Middle Ages, the Enlightenment and modern theories.[145] The natural rights theory considered individual rights derived from either the laws of God, nature or reason. It was incorporated into the American Declaration of Independence of 1776, whose second sentence reads, 'We hold these truths to be self-evident, that all men are created equal, that they are endowed by their Creator with certain unalienable Rights, that among these are Life, Liberty and the pursuit of Happiness'; and into the French Declaration of the Rights of Man and of Citizens of 1789, Article 2, which reads, 'The aim of all political association is the preservation of the natural and imprescriptible rights of man. These rights are liberty, property, security, and resistance to oppression.'

It is not my intention to discuss these theories in depth. The aim is to indicate the most relevant theories that could serve as a basis for later discussion about possible justifications for perceiving digital assets like property.[146] These theories could all be brought under the umbrella of three main groups, widely accepted and used, both in theory and practice. These theories for justifying private property are utilitarianism, the labour theory, and the

[142] For instance, Aristotle's definition of property provides that something 'is "our own" if it is in our power to dispose of it or keep it'. For more, see A Mossoff, 'What is property? Putting the pieces back together' (2003) 45 *Arizona Law Review* 371, 391. The difference between Plato and Aristotle was that the former was a communist when it came to property while the latter defended private property. See for example R Schlatter, *Private Property: The History of an Idea* (Allen & Unwin 1951) 9–21.

[143] Roman law scholar Barry Nicholas notes that there is no Roman definition of ownership, but there are Romanistic ones, usually emphasising enjoyment, and adapting usufruct by adding the right of abuse – *ius utendi fruendi abutendi*. In Mossoff (n 142) 391–2.

[144] Defined in Cicero's Republic or Institutes of Justinian. For more, see Schlatter (n 142) 21–33.

[145] For more, see A Ryan, *Property* (Open University Press 1987) 61–70.

[146] For a sound overview on some of these theories, concentrating mainly on the natural law ones, see S Buckle, *Natural Law and the Theory of Property: Grotius to Hume* (Clarendon Press 1991). For a discussion on Grotius', Pufendorf's and Locke's theories in relation to the bundle of rights and exclusive rights theory in common law scholarship, see Mossoff (n 142).

personhood theory (with the theory of human flourishing as, arguably, its subset).[147]

(a) Utilitarian Theories

The background assumptions of the utilitarian theory are that natural resources are scarce, people are demanding and of limited altruism, and labour is disagreeable and unpleasant. Therefore, there is a necessity to devise ways to make nature yield as much as it can, to be used in a way beneficial to the community. According to utilitarian theory, this aim cannot be achieved without property rules, since some will try to take the fruits of other people's efforts if they are not prevented. However, in the view of some authors, this argument does not readily distinguish between the need for property and the need for private property, since it is not certain whether, for example, capital would be best served in the hands of families or individuals.[148]

The principal exponent of the utilitarian justification of property, Jeremy Bentham, continued on the path traced earlier by Hume, who introduced the principle of utility, arguing that due to the selfish nature of people, and limited natural resources, it is helpful to respect the right of others in order to sustain our own and promote our happiness.[149]

In 1789, Bentham coined the term 'utilitarianism' and introduced the notion of a 'felicific calculus'.[150] This is the main principle of utilitarianism, looking for 'the greatest good for the greatest number'[151] and concentrating on the welfare approach to utility. This theory is, along with the labour theory, frequently used as justification for tangible and intangible property in the common law systems, especially in the US (by courts, academia and even the Constitution; see the discussion below).[152]

The underlying assumption is that people will create socially desirable objects only if granted appropriate incentives, some exclusive right, that is, property rights, so that 'free riders' do not enjoy the fruits of someone else's labour. Further, the theory assesses the goodness or badness of consequences regarding their tendency to maximise utility or welfare, where welfare maximises total net pleasure.

147 Becker, *Property Rights* (n 141) 99–101; Harris (n 3) 168.

148 See JS Mill and J Bentham, *Utilitarianism and Other Essays*, ed. A Ryan (Penguin 1987) 56.

149 See Schlatter (n 142) 239–42.

150 J Bentham, *An Introduction to the Principles of Morals and Legislation*, ed. JH Burns and HLA Hart (Athlone Press 1970) 12–13.

151 Ibid.

152 See also commentary in Alexander and Peñalver (n 140) 11.

For Bentham, the nature of utility is an individual, hedonistic pleasure. The problem that would arise in trying to maximise perverse pleasures, according to this theory, was solved by Mill, who classified them into higher and lower pleasures, giving more weight to higher pleasures when assessing decisions according to the felicific calculus. Hence his famous statement that 'It is better to be a human dissatisfied than a pig satisfied.'[153] If the decision concerns more than one person, the solution, according to Bentham, is to aggregate utility, to add up total net pleasure enjoyed by all the individuals in the group. This principle has been widely criticised on the grounds that it lacks sensitivity to unequal distribution,[154] therefore his principle would be satisfied if there is a total pleasure that only a tiny elite enjoys.

When applied to the justification of property, this theory argues that people need to individually acquire, possess, use and consume some things to achieve a reasonable degree of happiness. Security in possession and use of things is impossible unless enforced and unless modes of acquisition are controlled. The need for such control and enforcement amounts to the administration of a property rights system because of insecurity in possession and use and uncontrolled acquisition of the goods, which render individual achievement of a reasonable degree of happiness impossible. Therefore, a system of property rights is necessary.[155]

Based on the assumptions of the utilitarian theory, mainly in the US, the economic theory of property rights was developed. Many of these theorists refer to themselves as *wellfarists*, too. In response to the Benthamite individualist conception of pleasures, contemporary theorists adopted a weaker concept of goodness, the satisfaction of preferences, rather than pleasures in general. The idea is to filter for worthy and reasonable preferences, so that, in contrast to Bentham's conception, by concentrating on individual preferences, whatever they might be, later theorists tried to mitigate the absoluteness of pleasure, bringing it down to mere preferences, and only those worthy and reasonable, usually measured within a group rather than at the individual level. However, most of these theorists focus on the satisfaction of likely consequences rather than the actual consequence of the decision being evaluated.

In modern theory, the closest to Bentham's principle of aggregated utility is the Kaldor–Hicks criterion of efficiency.[156] According to this criterion, a social decision is superior to alternatives if the people who benefit from

[153] JS Mill, 'Utilitarianism' in Mill and Bentham (n 148) 260.

[154] See Alexander and Peñalver (n 140) 14.

[155] Becker, *Property Rights* (n 141) 57–8.

[156] K Nicholas, 'Welfare propositions in economics and interpersonal comparisons of utility' (1939) *Economic Journal* 49, 549–52.

the choice gain enough that they could, hypothetically, fully compensate those individuals who lose out from it such that the losers consider themselves no worse off than they were before. The Pareto principle states that a social choice is good if it makes at least one person better off without decreasing utility for anyone else.[157]

The biggest problem with assessing these principles is the gathering of necessary data. These methods vary from assessing the amount people are willing to pay to satisfy their preferences to conducting surveys and other empirical research. Thus, Alexander rightly argues that the value of any utilitarian prescription will only be as good as the empirical information on which it is based.[158]

Further, based on Benthamite assumptions, modern economic theories of property focus on explaining property in terms of economic factors, such as the efficiency achieved on the market, and are not normative like utilitarianism but rather explanatory.[159] In the best-known version of these theories, the inefficiency of common property is attributed to the tragedy of the commons. The 'tragedy of the commons' is a well-known and widely built upon concept in the US, created by Garrett Hardin.[160]

A tragedy of the commons is a situation appearing when too many owners have a privilege to use a resource, and no one has a right to exclude another. This leads to overuse and depletion of the resource. The theory is based on four assumptions: a community made up of rational actors who aim to maximise their individual material gain; a resource is rivalrous, that is, one who uses a resource progressively diminishes or degrades the remaining supply of the resource; users can keep all the benefits while they all bear the costs; and use of a resource is unregulated and open to all.[161]

On the other hand, as a critique, a theory of the 'tragedy of the anticommons' appeared, introduced by Michael Heller.[162] It is defined as a situation where 'multiple owners are each endowed with the right to exclude others from a scarce resource, and no one has an effective privilege of use. When too many owners hold rights of exclusion, the resource is prone to underuse – a tragedy of the anticommons.'[163] The example he uses is empty storefronts in transitional Moscow instead of metal kiosks appearing widely

[157] Alexander and Peñalver (n 140) 14.
[158] Ibid. 15. See also Becker, *Property Rights* (n 141) 68.
[159] For more, see Mill and Bentham (n 148) 103–15.
[160] G Hardin, 'The tragedy of the commons' (1968) 162 *Science* 1243, 1243–8.
[161] Alexander and Peñalver (n 140) 28.
[162] MA Heller, 'The tragedy of the anticommons: Property in the transition from Marx to markets reviewed' (1998) 111(3) *Harvard Law Review* 621, 621–88.
[163] Ibid. 624.

in the same areas. He argues that the underuse in this example is caused by an inappropriate initial endowment of property rights, where multiple owners were assigned rights to exclude others, and no one had a right to control the resource individually. Thus, he suggests that bundles of rights should be allocated to individuals rather than allocating individual rights, to enable more efficient control of a resource. Another example is IP rights, where competing and restrictive patent rights in biomedical and pharmaceutical research, for instance, could disable the introduction of useful and cheaper products to the market.[164]

A similar theory to that of the tragedy of the commons was put forward by Demsetz, based on Blackstone's theory. It describes the invention of property as a response to scarcity. He illustrates his theory through the case of Native American hunters in the Hudson Bay area during the early colonial period when the Europeans' demand for furs created immense pressure to capture furred animals. Consequently, the Native Americans began to overhunt the common grounds, each hunter imposing 'external' costs on the others since all needed to deploy more hunting effort to catch the rare animals. Therefore, according to Demsetz, these indigenous hunters realised they could prevent overhunting by turning their common hunting grounds into private property. Once they had done so, he recounts, the individual owners appropriated wildlife resources on their respective territories, and the private hunting grounds became productive again. In Demsetz's more technical economic terms, property rights enabled people to 'internalise externalities'.[165]

The utilitarian justification is, understandably, and bearing in mind the dominant legal and political culture, most accepted in the US and countries that embrace liberal capitalist values. Thus, even the US Constitution in Article I, section 8, clause 8 says that the purpose of protecting IP rights is 'to promote the Progress of Science and useful Arts',[166] which is a utilitarian concept. Moreover, the US courts often use this justification in deciding IP-related cases.[167] This theory will be employed in the following chapters, where the utility of recognising digital assets as property will be assessed.

[164] MA Heller and R Eisenberg, 'Can patents deter innovation? The anticommons in biomedical research' (1998) 280 *Science* 698, 698–701.

[165] H Demsetz, 'Toward a theory of property rights' (1967) 57 *American Economic Review* 347, 347–9.

[166] This clause is known as the Patent and Copyright Clause. For more, see I Donner, 'The copyright clause of the U.S. Constitution: Why did the framers include it with unanimous approval?' (1992) 36(3) *American Journal of Legal History* 361, 361–78.

[167] See for example *Mazer* v *Stein*, 347 US 201, 219 (1954); *New York Times Co* v *Tasini*, 533 US 483, 495 (2001); *Precision Instrument Mfg Co* v *Auto Maint Mach Co*, 324 US 806, 816 (1945).

(b) Labour Theories

Labour theory was introduced and developed by John Locke. Locke has had the most significant influence on property theories in English-speaking countries, and his theory is often associated with libertarians.[168] His theory is mainly elaborated in chapter five of the *Second Treatise of Government*. In the introduction to the 1980 edition of the *Second Treatise*, Macpherson observes that nobody has made a more persuasive case for unlimited property rights than Locke, who defended the limited constitutional liberal state, and asserts that ‘no one has come even near his skill in moving from a limited and equal to an unlimited and unequal property right by invoking rationality and consent’ and thus providing a justification for a liberal capitalist state conception of property.[169]

Locke’s central argument is that ‘whatsoever (man) removes out of the state that nature hath provided and left it in, he hath mixed his labour with, and joined to it something that is his own, and thereby makes it his property’.[170] Before asserting this, Locke starts by elaborating his intention to show how men could have property in something that ‘God gave mankind in common’ and all that without any explicit consent from others, the commoners.[171] Further, he claims that to use a natural resource (in his example fruit or venison) for ‘the best advantage of life, and convenience’, it is necessary to appropriate it ‘that another can no longer have any right to it’.[172] The justification for appropriation lies in Locke’s claim that everyone has property in his or her own person, consequently in their labour, ‘the work of his hands’. Therefore, whatever one removes from the commons and mixes their labour with ceases to be held in common and becomes their property.[173] This argument is not, however, without limitations.

Locke’s famous principle of waste limitation is the requirement that the labourer is limited to ‘as much as anyone can make use of to any advantage of life before it spoils’.[174] Becker defines it as the ‘why not?’ argument, meaning that a labourer has property rights if he produces something by his labour, and it is not against moral requirements for the labourer, and finally, other

[168] Alexander and Peñalver (n 140) 35.
[169] CB Macpherson, ‘Editor’s introduction’ in J Locke, *Second Treatise of Government: Essay Concerning the True Original Extent and End of Civil Government*, ed. with an intro. CB Macpherson (Hackett 1980) xxi.
[170] Locke (n 169) § 27, 19.
[171] Ibid. § 25, 18.
[172] Ibid. § 26, 19.
[173] Ibid. § 27.
[174] Ibid. § 56.

members of society do not incur loss from being excluded from enjoying the fruit of labour.[175]

The final limitation is Locke's 'enough and as good' proviso, meaning that there should be enough and as good left in common for others after appropriation. Note, however, that Locke revises these limitations when writing of the introduction of money as property when all three limitations were removed – spoilage, 'enough and as good' and mixing labour – justifying unlimited private property.[176] Money does not spoil, so the first limitation is removed.[177] The second limitation is abandoned with the development of commerce and the consent to use money. Locke explains it by referring to the value of appropriated and cultivated land, which is at least ten times more than that of unappropriated and uncultivated land, and thus, even if there is not enough land left for others, there is *produce* for everyone.[178] The advent of commerce then induced each man to appropriate more and exchange the surplus for money, 'beyond the use of his family'.[179]

Penner criticises Locke's theory because it does not make a real difference between contracts and property based on consent. Moreover, according to him, the theory is flawed since it presupposes one's right to control or the ability to act freely, which is not always the case (the problem of slavery or ownership of goods created by employees).[180]

Nozick offered one of the most famous critiques of Locke's theory. It is interestingly summarised in this widely cited quote:

> Why isn't mixing what I own with what I don't own a way of losing what I own rather than a way of gaining what I don't? If I own a can of tomato juice and spill it in the sea so that its molecules (made radioactive, so I can check this) mingle evenly throughout the sea, do I thereby come to own the sea, or have I foolishly dissipated my tomato juice?[181]

Becker, for instance, shares this view,[182] whereas Alexander responds to this objection using the example of a stream, soup and tomato juice. Contrary to Nozick's example of the sea, in Alexander's illustration there is only a little water taken from the stream, so tomato juice adds much more value to the

175 Becker, 'Moral basis of property rights' (n 18) 193.
176 Macpherson, 'Editor's introduction' (n 169) xvii.
177 Locke (n 169) § 37.
178 Ibid. § 37; commentary in Macpherson, 'Editor's introduction' (n 169) xvii.
179 Locke (n 169) § 48.
180 Penner, *Idea of Property* (n 58) 187–8.
181 R Nozick, *Anarchy, State and Utopia* (Basil Blackwell 1974) 175.
182 Becker, *Property Rights* (n 141) 34.

water in making a soup. A further objection Alexander makes is collective labour, which is usually how things are produced.[183]

Another problem with Locke's theory is its application when the common is already appropriated. Throughout history, it has been used as a justification for various kinds of property objects, but many authors disregarded the fact that the theory applies only to the justification for things appropriated from the commons.[184] Locke's primary goal was to justify the appropriation of Native American land. He made it quite clear that he spoke of nature's state and not of his contemporary society.[185] This can be seen in his example of the commons in England, where the common is created by 'the law of the land' and to be used by all countrymen, who already have their property and participate in commerce. Hence, no one can appropriate this land without consent as it is 'in the beginning and first peopling of the great common of the world'.[186] This objection is particularly relevant to the discussion on digital assets, as it is challenging to identify the relevant commons there.

As hinted above, Locke's work has been used by a variety of scholars in efforts to justify diverse kinds of property relations, institutions and objects. The arguments supported by this theory have been used by opposing theories and practices. Thus, the nineteenth-century reading of Locke supported egalitarian and redistributive efforts and policies, whereas, conversely, the twentieth-century readings paint a picture of Locke as a defender of capitalist accumulation and rights of property.[187]

Locke's theory has been used as a starting point and an essential reference in discussions on IP and information as property (see 'Property in emails' in Chapter 6). This book will use it as one of the theories that might justify property in certain digital assets.

[183] Alexander and Peñalver (n 140) 48.

[184] Ibid. 37.

[185] Locke (n 169) § 51. See also Schlatter (n 142) 156.

[186] Locke (n 169) § 35, 22.

[187] See CB Macpherson, *The Political Theory of Possessive Individualism: Hobbes to Locke* (Wynford edn, new edn, Oxford University Press 2011); Alexander and Peñalver (n 140) 53; and Nozick and Epstein as the most influential contemporary American libertarian thinkers: Nozick (n 181) and R Epstein, 'One step beyond Nozick's minimal state: The role of forced exchanges in political theory' (2005) 22 *Social Philosophy and Policy* 286. See J Tully, *A Discourse on Property: John Locke and His Adversaries* (Cambridge University Press 1980) 172. Becker, *Property Rights* (n 141) 43. Becker asserts that Locke's theory provides more justification for socialist than capitalist societies.

(c) Personality Theory

Personality (personhood) theories justifying property originated from Hegel's conception of property as an extension of personality.[188] It is 'the relation of personality to the external sphere of things, understood in terms of the free will'.[189] Also, 'A person must translate his freedom into an external sphere in order to exist as an idea.'[190] Further, free will in every stage of development is embodied in something externalised, and since things have no will, there is an absolute right of appropriation. Therefore, a person with a will has a right to appropriate and determine the use of a thing that is considered as not having a will. This relates to the first appropriation. The following person who wants to appropriate it is confronted by the will of another embodied in possession of a thing.[191] Hegel, however, did not specify what types of property should exist, which is a lack in his theory.[192]

Modern personhood theories mainly focus on personality and human rights, invoking autonomy, liberty, identity and privacy as values that could not be effectively protected without property rights; in other words, these values are intrinsically connected to property interests. Thus, as Munzer notes, personality theorists advocate private property rights that should be recognised to advance a broad range of interests and promote human flourishing. He summarises these interests as peace of mind, privacy, self-reliance, self-realisation (as a social being and as an individual), security and leisure, responsibility, identity, citizenship and benevolence.[193]

In the cluster of modern personhood theories, one of the most significant and influential is Radin's personhood theory.[194] Radin classifies property as fungible and personal, arguing that property is an essential vehicle for the development of personality and, therefore, that the property which is incredibly close to a person's self-definition (e.g. his home, a wedding ring) deserves special legal protections and precedence over other property rights.[195] Critics,

[188] GWF Hegel, *The Philosophy of Right*, trans. TM Knox (Oxford University Press 1967). For a sound elaboration and critique, see Penner, *Idea of Property* (n 58) 169–86.

[189] See Penner, *Idea of Property* (n 58) 173.

[190] Hegel (n 188) para 41.

[191] Penner, *Idea of Property* (n 58) 174. For more, see Alexander and Peñalver (n 140) 59–61.

[192] Alexander and Peñalver (n 140) 65.

[193] SR Munzer (ed.), *New Essays in the Legal and Political Theory of Property* (Cambridge University Press 2001) 189–90; here he refers to personhood theorists, Waldron, Fried, Rose, Radin, Green etc.

[194] MJ Radin, 'Property and personhood' (1982) 34 *Stanford Law Review* 957; Waldron (n 12).

[195] Radin (n 194).

however, suggest that the shortcoming of her theory is that Radin only referred to the autonomous self, individual development, and had nothing to say about the relational, interpersonal aspect of property and self-development.[196]

This theory is also an essential underpinning of the discussions on whether information and digital assets can be considered property. In addition, it is closely linked to theories of autonomy and privacy, explored in Chapter 3, and it will be referred to further there.

2.4 Concluding Remarks

In conclusion, recognising an object or phenomenon as property brings some general benefits to the rights holder. These advantages include the *in rem* effect against the whole world and not only between parties to a contract; remedies applicable to property; and, importantly, the possibility to transmit it on death (with the reservations discussed above).[197] Regarding the last element, if an environment is extensively regulated by contracts (e.g. terms of service of Internet developers and service providers), which are strictly personal or explicitly forbid transmission on death, according to the rules mentioned above, they cannot be transmitted on death. Therefore, if their social and economic value is such that transmission on death is desirable, the property regime would be more suitable. This is, nevertheless, a provisional conclusion, dependent on the findings in the following chapters, where the nature and value of individual digital assets are discussed. Each chapter will either confirm or reject the proposition that digital assets can be considered property and therefore generally transmitted on death like any other form of property.

We could provisionally conclude that new forms of property could fit more easily into some of the English categories (choses in action), founded on the bundle of rights theory of property. However, notwithstanding the rigid 1885 (in *Colonial Bank* v *Whinney*) categorisation of personal property in English law, many authors would argue that, unlike in civil law, common law property is capable of expansion and inclusion of new categories.[198] Lord Wilberforce supports this in the *National Provincial Bank Ltd* v *Ainsworth* statement, cited in the previous section. Also, as Ball rightly notes, lack of a principle of unity of property in English law, lack of limitative definitions of

[196] See Alexander and Peñalver (n 140) 69.

[197] See for example Honoré (n 5); Becker, 'Moral basis of property rights' (n 18); *OBG Ltd* v *Allan* [2007] UKHL 21; [2008] 1 AC 1, 309.

[198] WG Friedmann, *Law in a Changing Society* (2nd edn, Penguin 1972) 94; K Moon, 'The nature of computer programs: tangible? Goods? Personal property? Intellectual property?' (2009) *European Intellectual Property Review* 396, 407.

property, and the bundle of rights conception of property make English law liberal and prone to fragmentation and the manipulation of property rights by lawyers.[199]

Nevertheless, nowadays, theory no longer reflects reality as the courts have refused to create new forms of property in the last century.[200] In addition to the English example of confidential information, discussed in more detail later in Chapter 6, there is electricity. Even legal recognition of full property rights for choses in action, recognised as property in English law (see the discussion earlier in this section), has been denied because this property is not tangible. Thus, in *OBG Ltd* v *Allan*,[201] the House of Lords, by a 3:2 majority, denied applying the tort of conversion to anything other than chattels. However, this is not the case in the US, where the courts in some states have abandoned this traditional view, and some information (e.g. fresh news, trade secrets) is considered property. This development will be discussed further in Chapter 6.

Theories of property presented in this chapter will be examined systematically in each case study chapter to analyse whether doctrinal law supports the property status of digital assets and whether the theory, policy and ethics suggest that property should be recognised in such assets.

199 Ball (n 59) 4.

200 '[E]lectricity . . . is not capable of ownership' in *Low* v *Blease* [1975] Crim LR 513. See commentary in Moon (n 198) 406–7.

201 [2007] UKHL 21.